

# Scan-Line Algorithm.

(1) Draw the polygon.

First draw the polygon using the given vertices.

(2) Construct the Edge Table.

For every edge :

ignore horizontal edges.

Find : 

|           |              |       |
|-----------|--------------|-------|
| $y_{max}$ | $x(y_{min})$ | $1/m$ |
|-----------|--------------|-------|

store them in Edge Table.

(3) Start with the lowest scan.

(3) Insert Edges into Active Edge Table.  
and Find the intersecting points

(4) Remove the completed Edges.

(5) After filling one scan line, move to the next scan line by updating the intersection x-value using

$$x_{new} = x_{old} + \frac{1}{m}$$

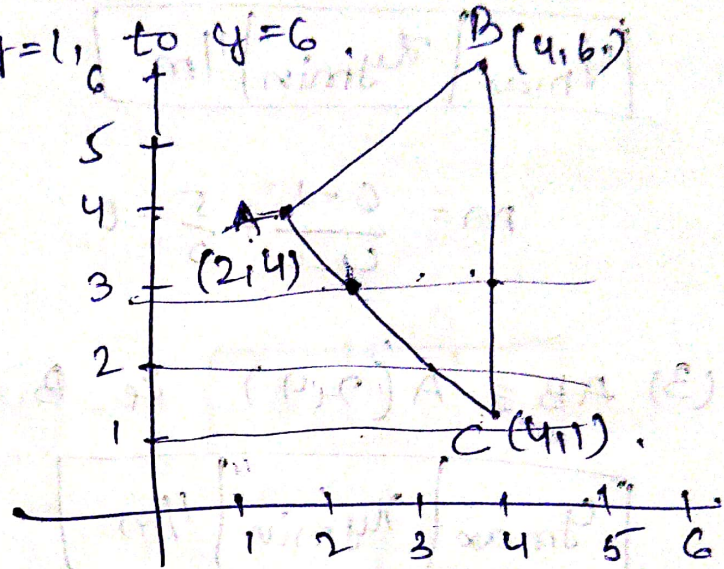
(6) When there are no more edges in both the Edge Table and Active Edge Table, the entire polygon has been filled.

→ Determine the content of the AET to fill the polygon with vertices  $A(2,4)$ ,  $B(4,6)$ ,  $C(4,1)$  for  $y=1$  to  $y=6$ .

Sort the edges from

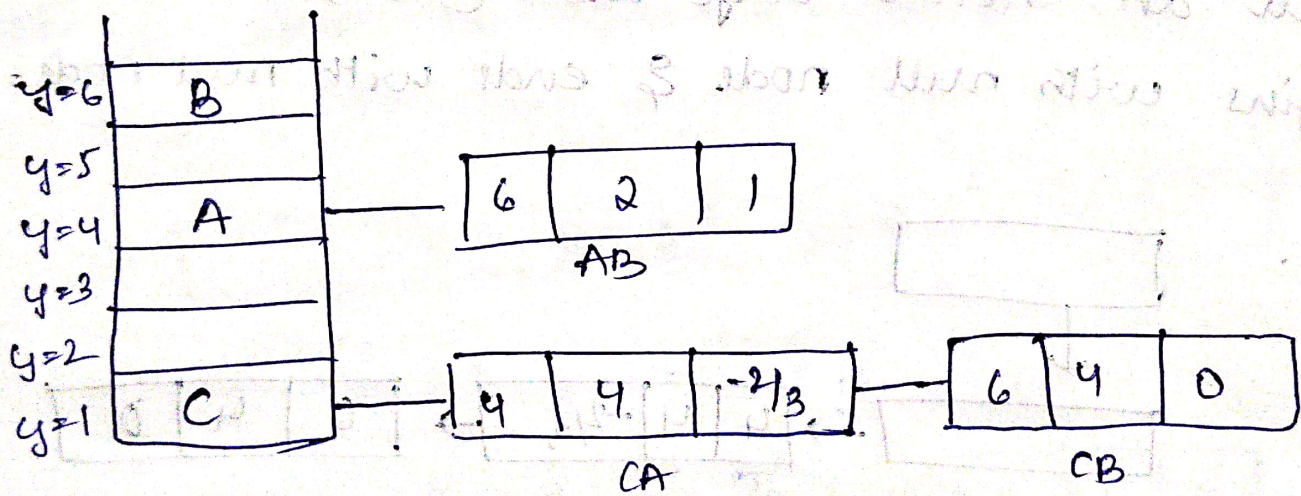
$y_{\min} \rightarrow y_{\max}$ .

$C(4,1)$ ,  $A(2,4)$ ,  $B(4,6)$ .



Create a Global Edge table (GET).

It is an array of linked list.



$C \rightarrow CA, CB$ .

(1)  $CA = C(4,1)$ ,  $A(2,4)$

| $y_{\max}$ | $x_{y_{\min}}$ | $1/m$ |
|------------|----------------|-------|
|------------|----------------|-------|

|   |   |                |
|---|---|----------------|
| 4 | 4 | $-\frac{2}{3}$ |
|---|---|----------------|

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 - 1}{2 - 4} = \frac{3}{-2}$$

(2)  $CB \Rightarrow C(4,1)$  ,  $B = (4,6)$ .

|           |            |       |
|-----------|------------|-------|
| $y_{max}$ | $x_{ymin}$ | $1/m$ |
|-----------|------------|-------|

|   |   |   |
|---|---|---|
| 6 | 4 | 0 |
|---|---|---|

$$m = \frac{6-1}{4-4} = \frac{5}{0} = \infty$$

(3)  $AB = A(2,4)$  &  $B(4,6)$ .

|           |            |       |
|-----------|------------|-------|
| $y_{max}$ | $x_{ymin}$ | $1/m$ |
|-----------|------------|-------|

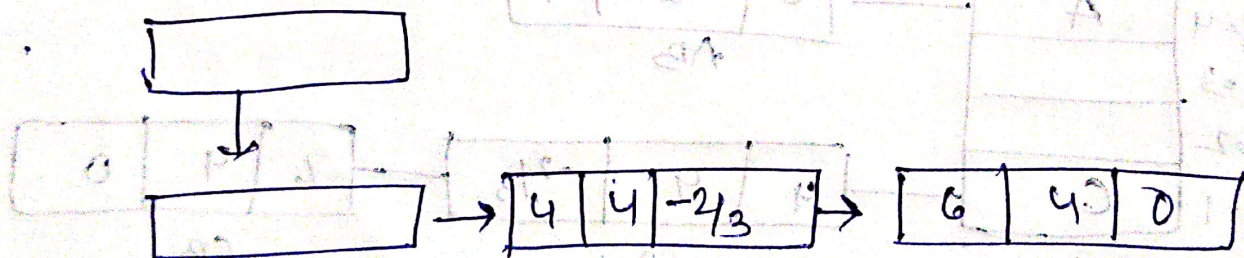
|   |   |   |
|---|---|---|
| 6 | 2 | 1 |
|---|---|---|

$$m = \frac{6-4}{4-2} = \frac{2}{2} = 1$$

Create an Active Edge table (AET). —

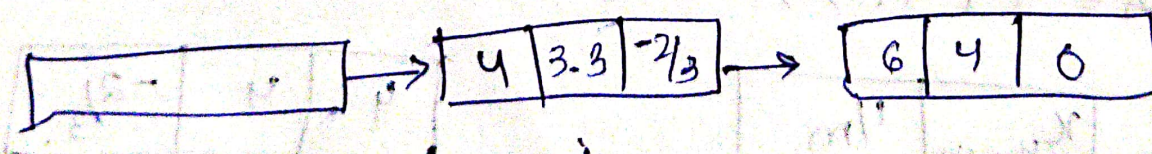
Begins with null node & ends with null node.

$i=1$ .



$\{(4,1), (4,1)\}$ .

$i=2$



Increment  $x$  with the slope.

$$4 - \frac{2}{3} = \frac{10}{3} = 3.3$$

$\{(3.3, 2), (4, 2)\}$

$$i=3, \quad \boxed{\phantom{0}} \boxed{\phantom{0}} \boxed{\phantom{0}} \rightarrow \boxed{4} \boxed{\frac{8}{3}} \boxed{-\frac{2}{3}} \rightarrow \boxed{6} \boxed{4} \boxed{0}$$

$$\frac{8}{3} - \frac{2}{3} = \frac{8}{3} = 2.6$$

$$\{(2, 6, 3), (4, 3)\}$$

$$i=4, \quad \boxed{\phantom{0}} \boxed{\phantom{0}} \boxed{\phantom{0}} \rightarrow \boxed{4} \boxed{\frac{8}{3}} \boxed{-\frac{2}{3}} \rightarrow \boxed{6} \boxed{4} \boxed{0}$$

$$\frac{8}{3} - \frac{2}{3} = \frac{6}{3} = 2 \rightarrow \boxed{6} \boxed{2} \boxed{1}$$

$$\{(2, 4), (4, 4)\}$$

with 3 nodes we cannot make a pair.  
so delete the node whose first value is same.

i-value is same:

$$\Rightarrow \boxed{\phantom{0}} \boxed{\phantom{0}} \boxed{\phantom{0}} \rightarrow \boxed{6} \boxed{2} \boxed{1} \rightarrow \boxed{6} \boxed{4} \boxed{0}$$

$$\{(2, 4), (4, 4)\}$$

$$i=5, \quad \boxed{\phantom{0}} \boxed{\phantom{0}} \boxed{\phantom{0}} \rightarrow \boxed{6} \boxed{3} \boxed{1} \rightarrow \boxed{6} \boxed{4} \boxed{0}$$

$$2 + 1 = 3$$

$$\{(3, 5), (4, 5)\}$$

$$i=6, \quad \boxed{\phantom{0}} \boxed{\phantom{0}} \boxed{\phantom{0}} \rightarrow \boxed{6} \boxed{4} \boxed{1} \rightarrow \boxed{6} \boxed{4} \boxed{0}$$

$$\{(4, 6), (4, 6)\}$$

## Edge Table

| Edge | $y_{max}$ | $x_{ymin}$ | $1/m$  |
|------|-----------|------------|--------|
| CA   | 4         | 4          | $-2/3$ |
| CB   | 6         | 4          | 0      |
| AB   | 6         | 2          | 1      |

## Active Edge Table

| Scan line (y) | Active Edges | x intersections | Pixels Filled         |
|---------------|--------------|-----------------|-----------------------|
| 1             | CA, CB       | 4, 4            | <del>(4,1)</del> None |
| 2             | CA, CB       | 3.33, 4         | (4,2)                 |
| 3             | CA, CB       | 2, 6, 4         | (3,3), (4,3)          |
| 4             | AB, CB       | 2, 4            | (2,4), (3,4)<br>(4,4) |
| 5             | AB, CB       | 3, 4            | (3,5), (4,5)          |
| 6             | None         | —               | None                  |